

Tiny Bodies, Big Brains — Module 01: Systems — Think About It

Part A: Family Discussion Questions

1. Why do you get hiccups? (Your diaphragm muscle is twitching — a body system glitch!).
2. What does your body need every day to run well? (Food, water, air, sleep, love).
3. Why does your heart beat faster when you run? (The engine works harder to deliver fuel to your muscles!).
4. What happens when you eat too much candy? (The system gets a sugar overload and then crashes).
5. Why do you need sleep? (The body repairs itself and the brain does upgrades).
6. How is your body like a car? (It has an engine, fuel, wheels, and a computer).
7. What is the biggest system inside your body? (Your skeleton — 206 bones holding everything up!).
8. Why do you shiver when you are cold? (Your body is trying to warm up — homeostasis!).
9. Can you name three "sub-systems" inside your body? (Digestion, breathing, circulation).
10. What would happen if your stomach stopped working but your legs kept running? (You would run out of energy! Systems depend on each other).

Part B: Hands-On Activities and Reflection

1. Find your pulse on your wrist or neck. Count the beats for 15 seconds and multiply by 4. That is your resting heart rate! Now do 20 jumping jacks and check again. What happened?
2. Lie on a big piece of paper. Have someone trace your body outline. Draw the systems inside: heart (red), lungs (blue), stomach (green), brain (yellow). Label everything!

3. Breathe in and count how long you can hold your breath. (Be safe! Stop when it gets uncomfortable). Your body MADE you breathe — that is homeostasis protecting the system.
4. Track what you eat and drink for one meal. List the "inputs." Then list what the body does with them ("Bread gives me energy," "Water keeps me cool"). You are fueling your system!
5. Measure your height and mark it on a wall or doorframe. Compare it to a mark from a year ago (or start one now!). Growth is the system upgrading itself.
6. Feel your forehead. Is it warm? Your body keeps itself at about 98.6 degrees. Put a cold cloth on your arm. What does the body do? (Sends warm blood to heat it up!).
7. Make a "Body Owner's Manual" — a booklet with a page for each system (heart, brain, stomach, muscles). Draw each one and write one care tip ("Feed me vegetables!").
8. Try to wiggle your ears, curl your tongue, or touch your nose with your tongue. Not everyone can! Our body systems are unique. What can YOUR body do?
9. Listen to your stomach. Is it quiet or gurgling? That sound is digestion at work! Your body is processing fuel right now. Write down what you think is happening inside.
10. Before bed, put your hand on your chest and feel your breathing slow down. Thank your body systems for working all day long. They never take a break!